PARENT ONLY — SOLVED ANSWER KEY

Vyom · Parent answer key

15 Feb – 21 Feb 2027 · Monday–Sunday · NDPS Class 9

Vyom ko test se pehle ye PDF mat dena. Student pack mein sirf questions aur blank answer space hai. Is alag parent PDF mein har original question ke saath uska matching solved answer, working aur marking points diye hain; Test ID se student paper match karein.

Checking: same evening. C = concept, R = recall, K = careless, T = time.

90%+: short revision. 75–89%: weak-topic practice + retest. Below 75%: topic dobara samjhao; mistakes notebook mein likho.

Answer keys existing source question banks se li gayi hain; school pattern abhi unconfirmed hai. Alternate valid answers ko marking guidance ke mutabik accept karein.

Week score register

Date / test ID	Test	Marks	Galti / retest
15 Feb T2027-02-15	Hindi Mock (2-hr section practice)	___ / 80	Retest: 18 Feb
16 Feb T2027-02-16	Maths Mock (2-hr section practice)	___ / 80	Retest: 19 Feb
17 Feb T2027-02-17	Science Mock (2-hr section practice)	___ / 80	Retest: 20 Feb
18 Feb T2027-02-18	SST Mock (2-hr section practice)	___ / 80	Retest: 21 Feb
19 Feb T2027-02-19	English Mock (2-hr section practice)	___ / 80	Retest: 22 Feb
20 Feb T2027-02-20	AI Mock (2-hr section practice)	___ / 50	Retest: 23 Feb

हिंदी · मॉक प्रश्न-पत्र 1 · Answer key

Monday, 15 February 2027 · 80 marks

Test ID: T2027-02-15

कक्षा 9 · गंगा (पाठ 1–12) + व्याकरण + लेखन — पूरा पाठ्यक्रम

Q1 [7]

Question: निम्नलिखित गद्यांश पढ़कर प्रश्नों के उत्तर दीजिए —

> पानी की हर बूँद कीमती है। हमारे देश के कई हिस्सों में गर्मियों में लोगों को कई किलोमीटर चलकर पानी लाना पड़ता है, जबकि शहरों में नल खुले छोड़ देना आम बात है। पुराने समय में गाँवों में बावड़ियाँ, तालाब और कुएँ बनाए जाते थे, जिनमें वर्षा का जल इकट्ठा होता था। लोग इन्हें साफ़ रखने को अपना धर्म मानते थे।
> आज भू-जल का स्तर लगातार नीचे जा रहा है। यदि हम छत पर गिरने वाले वर्षा-जल को टैंकों या ज़मीन में पहुँचा दें, तो भू-जल फिर से भर सकता है। ब्रश करते समय नल बंद रखना, गाड़ी को बाल्टी से धोना और रिसते नलों को ठीक करवाना जैसे छोटे कदम भी बड़ा अंतर ला सकते हैं। जल बचाना केवल सरकार का नहीं, हम सबका दायित्व है।

(i) गर्मियों में कई लोगों को क्या कठिनाई होती है? [1]
(ii) पुराने समय में वर्षा-जल कहाँ इकट्ठा किया जाता था? [1]
(iii) भू-जल को फिर से भरने का कौन-सा उपाय बताया गया है? [1]
(iv) जल बचाने के दो छोटे कदम लिखिए। [2]
(v) “दायित्व” शब्द का अर्थ लिखिए। [1]
(vi) गद्यांश का उपयुक्त शीर्षक लिखिए। [1]

Expected answer / marking points: (i) कई किलोमीटर चलकर पानी लाना [1]। (ii) बावड़ियों, तालाबों और कुओं में [1]। (iii) छत के वर्षा-जल को टैंकों/ज़मीन में पहुँचाना [1]। (iv) ब्रश करते समय नल बंद रखना; बाल्टी से गाड़ी धोना; रिसते नल ठीक करवाना (कोई दो) [2]। (v) ज़िम्मेदारी/कर्तव्य [1]। (vi) “जल ही जीवन है” / “बूँद-बूँद की बचत” (कोई उपयुक्त) [1]।

Q2 [7]

Question: निम्नलिखित काव्यांश पढ़कर प्रश्नों के उत्तर दीजिए —

> सुबह-सुबह जब सूरज जागे, खेतों में सोना बिखराए,
> किसान चले हल लेकर अपने, धरती माँ के गीत सुनाए।
> पसीने की हर बूँद से उसकी, फसलें लहर-लहर मुसकाएँ,
> भूखे को रोटी देने का, सपना वह हर दिन दोहराए।
> धूप सहे, बरसात सहे वह, फिर भी मन में आस जगाए,
> ऐसे अन्नदाता के आगे, सारा देश शीश झुकाए।

(i) काव्यांश में किसका वर्णन है? [1]
(ii) सूरज खेतों में क्या बिखराता है? [1]
(iii) किसान का सपना क्या है? [1]
(iv) किसान किन कठिनाइयों को सहता है? [1]
(v) “अन्नदाता” का अर्थ लिखिए। [1]
(vi) काव्यांश का केंद्रीय भाव दो वाक्यों में लिखिए। [2]

Expected answer / marking points: (i) किसान का [1]। (ii) सोना (सुनहरी धूप) [1]। (iii) भूखे को रोटी देना [1]। (iv) धूप और बरसात [1]। (v) अन्न देने वाला — किसान [1]। (vi) किसान कठिन परिश्रम करके देश का पेट भरता है; उसके प्रति हमें सम्मान रखना चाहिए [2]।

Q3 [2]

Question: उपसर्ग अलग कीजिए — (क) प्रचार (ख) दुर्भाग्य

Expected answer / marking points: (क) प्र + चार [1] (ख) दुर् + भाग्य [1]।

Q4 [2]

Question: प्रत्यय अलग कीजिए — (क) सुंदरता (ख) लड़ाकू

Expected answer / marking points: (क) सुंदर + ता [1] (ख) लड़ + आकू [1]।

Q5 [2]

Question: अर्थ के आधार पर वाक्य-भेद लिखिए — (क) कृपया शांत रहिए। (ख) वाह! क्या सुंदर दृश्य है!

Expected answer / marking points: (क) आज्ञावाचक/विधानवाचक (आज्ञा) वाक्य [1] (ख) विस्मयादिबोधक वाक्य [1]।

Q6 [2]

Question: रचना के आधार पर वाक्य-भेद लिखिए — (क) मैंने एक लड़का देखा जो बहुत तेज़ दौड़ता था। (ख) सूरज निकला और अँधेरा मिट गया।

Expected answer / marking points: (क) मिश्र वाक्य [1] (ख) संयुक्त वाक्य [1]।

Q7 [2]

Question: अलंकार पहचानिए — (क) रघुपति राघव राजा राम। (ख) सिंधु-सा विस्तृत और अथाह मन।

Expected answer / marking points: (क) अनुप्रास — "र" वर्ण की आवृत्ति [1] (ख) उपमा — मन की तुलना सिंधु से, वाचक "सा" [1]।

Q8 [2]

Question: वाच्य बदलिए — (क) लता जी ने अनेक गीत गाए। (ख) बच्चों से दौड़ा नहीं जाता।

Expected answer / marking points: (क) लता जी द्वारा अनेक गीत गाए गए [1] (ख) बच्चे दौड़ नहीं पाते [1]।

Q9 [2]

Question: रेखांकित पदों का परिचय दीजिए — "हीरा और मोती बहुत समझदार थे।"

Expected answer / marking points: हीरा — व्यक्तिवाचक संज्ञा, पुल्लिंग, एकवचन, कर्ता [1]; बहुत — प्रविशेषण/परिमाणवाचक क्रियाविशेषण ("समझदार" की विशेषता) [1]।

Q10 [2]

Question: संधि-विच्छेद या समास (जो पाठ्यक्रम में हो) — (क) "विद्यालय" का संधि-विच्छेद (ख) "देश-प्रेम" का समास-विग्रह

Expected answer / marking points: (क) विद्या + आलय [1] (ख) देश के लिए प्रेम/देश का प्रेम — तत्पुरुष [1]।

Q11 [5]

Question: पाठ 1 "दो बैलों की कथा" पाठ्यपुस्तक में खोलिए और वह अंश पढ़िए जहाँ हीरा-मोती काँजीहौस (मवेशीखाने) में बंद होते हैं। फिर उत्तर दीजिए —
(क) काँजीहौस में जानवरों की क्या दशा थी? [2]
(ख) हीरा-मोती ने वहाँ क्या साहसिक कार्य किया? [2]
(ग) इससे उनके चरित्र की कौन-सी विशेषता प्रकट होती है? [1]

Expected answer / marking points: (क) जानवर भूखे-प्यासे, कमज़ोर और बंधे हुए थे (पाठ के अनुसार) [2]। (ख) दीवार तोड़कर दूसरे जानवरों को आज़ाद कर दिया — हीरा ने सींगों से दीवार तोड़नी शुरू की, मोती ने उसे गिरा दिया; हीरा के बँध जाने पर मोती मित्र को छोड़कर नहीं गया [2]। (ग) परोपकार, साहस और मित्रता [1]।

Q12 [5]

Question: निम्नलिखित पद पढ़कर प्रश्नों के उत्तर दीजिए —

> प्रभु जी तुम दीपक, हम बाती, जाकी जोति बरै दिन राती।
> प्रभु जी तुम मोती, हम धागा, जैसे सोनहिं मिलत सुहागा।

(क) कवि ने प्रभु को क्या-क्या कहा है? [1]
(ख) "दीपक और बाती" के उदाहरण से कवि क्या कहना चाहते हैं? [2]
(ग) "सोनहिं मिलत सुहागा" का भाव लिखिए। [1]
(घ) इन पंक्तियों में कौन-सा अलंकार है? [1]

Expected answer / marking points: (क) दीपक और मोती [1]। (ख) जैसे बाती दीपक में जलकर दिन-रात प्रकाश देती है, वैसे ही भक्त प्रभु के साथ जुड़कर उनकी भक्ति में लगा रहता है; दोनों अलग नहीं [2]। (ग) जैसे सुहागा सोने से मिलकर उसकी चमक बढ़ाता है, वैसे ही भक्त प्रभु से मिलकर शोभा पाता है [1]। (घ) रूपक (और उपमा "जैसे") [1]।

Q13 [2]

Question: "क्या लिखूँ" में लेखक ने निबंध लिखने की कठिनाई क्यों बताई है?

Expected answer / marking points: लेखक को वह मानसिक स्थिति नहीं मिलती जिसमें भाव अपने आप उठें [1]; उन्हें सोचना और परिश्रम करना पड़ता है [1]।

Q14 [2]

Question: "संवादहीन" कहानी में ताई का अकेलापन किस प्रकार दिखाया गया है?

Expected answer / marking points: ताई के पास बात करने के लिए केवल मिट्ठू तोता है [1]; परिवार के लोग दूर हैं, इसलिए वह तोते से ही प्रेम और नोक-झोंक करती है [1]।

Q15 [2]

Question: "आखिरी चट्टान तक" में लेखक को कन्याकुमारी में कौन-से दृश्य सबसे अधिक प्रभावित करते हैं? (कोई दो)

Expected answer / marking points: सूर्योदय/सूर्यास्त के दृश्य [1]; तीन समुद्रों का संगम और विवेकानंद चट्टान (कोई दो) [1]।

Q16 [2]

Question: परशुराम के क्रोध के उत्तर में राम ने क्या कहा?

Expected answer / marking points: राम ने विनम्रता से कहा कि धनुष तोड़ने वाला आपका ही कोई दास होगा [1]; क्या आज्ञा है, मुझसे कहिए — इस प्रकार उन्होंने क्रोध शांत करने का प्रयास किया [1]।

Q17 [2]

Question: "भारति जय विजय करें" कविता में कवि भारत माता से क्या कामना करते हैं?

Expected answer / marking points: भारत की सदा जय और विजय हो [1]; देश की समृद्धि, सुंदरता और गौरव बना रहे [1]।

Q18 [2]

Question: "घर की याद" कविता में कवि को सबसे अधिक किसकी याद आती है और क्यों?

Expected answer / marking points: पिता (और माँ, भाई-बहनों) की [1]; जेल में बरसात के दिन घर-परिवार का प्रेम और पिता का उत्साही स्वभाव याद आता है [1]।

Q19 [4]

Question: "रीढ़ की हड्डी" एकांकी का शीर्षक कहाँ तक सार्थक है? उदाहरण सहित लिखिए। (60–80 शब्द)

Expected answer / marking points: शंकर का अपना कोई विचार नहीं, पिता की हाँ में हाँ मिलाता है — "बिना रीढ़ का" [2]; उमा का स्वाभिमान और साहस ही असली "रीढ़" है [1]; समाज की खोखली सोच पर व्यंग्य — शीर्षक सार्थक [1]।

Q20 [4]

Question: "झाँसी की रानी" कविता के आधार पर लक्ष्मीबाई के व्यक्तित्व की चार विशेषताएँ लिखिए। (60–80 शब्द)

Expected answer / marking points: वीरता, साहस, देशप्रेम, स्वाभिमान, कुशल योद्धा/नेतृत्व — उदाहरण सहित (कोई चार) [4]।

Q21 [5]

Question: किसी एक विषय पर 120 शब्दों में अनुच्छेद लिखिए —
(क) जल ही जीवन है (ख) मेरा प्रिय त्योहार (ग) किसान — हमारा अन्नदाता

Expected answer / marking points: विषय-वस्तु [2], भाषा-शुद्धता [1.5], क्रमबद्धता/प्रस्तुति [1.5]।

Q22 [5]

Question: अपने क्षेत्र में पानी की कमी की समस्या की ओर ध्यान दिलाते हुए नगर निगम अधिकारी को पत्र लिखिए।

ATHVA / OR

अपनी छोटी बहन को पत्र लिखकर समय पर सोने और जागने का महत्त्व समझाइए।

Expected answer / marking points: औपचारिक पत्र-प्रारूप [1.5]; समस्या, कारण, सुझाव [2.5]; समापन, भाषा [1]।
अथवा: अनौपचारिक पत्र-प्रारूप [1.5]; विषय-वस्तु [2.5]; भाषा [1]।

Q23 [5]

Question: परीक्षा की तैयारी को लेकर माँ और बेटे के बीच 80–100 शब्दों में संवाद लिखिए।

Expected answer / marking points: संवाद-प्रारूप व पात्रों के नाम [1]; स्वाभाविक, विषय से जुड़े संवाद [3]; भाषा [1]।

Q24 [5] **Question:** विद्यालय में “जल संरक्षण सप्ताह” के आयोजन की सूचना 40–50 शब्दों में लिखिए।

ATHVA / OR

अपने मित्र को उसकी जन्मदिन की बधाई देते हुए एक ई-मेल लिखिए (60–80 शब्द)।

Expected answer / marking points: सूचना-प्रारूप — विद्यालय का नाम, “सूचना”, दिनांक, शीर्षक [2]; विषय — क्या, कब, कहाँ, कौन [2]; नाम/पद [1]।
अथवा: ई-मेल प्रारूप — प्रति, विषय, संबोधन [2]; विषय-वस्तु [2]; समापन [1]।

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Maths · Mock Paper 1 · Answer key

Tuesday, 16 February 2027 · 80 marks

Test ID: T2027-02-16

Class 9 · poora syllabus (Ch 1–14)

Q1 [1]

Question: The coordinates of the origin are:

(a) (1, 1)
(b) (0, 0)
(c) (0, 1)
(d) (1, 0)

Expected answer / marking points: (b) (0, 0)

Q2 [1]

Question: The point (5, –2) lies in quadrant:

(a) I
(b) II
(c) III
(d) IV

Expected answer / marking points: (d) IV

Q3 [1]

Question: If $p(x) = 2x + 5$, then $p(-1)$ =

(a) 3
(b) 7
(c) –3
(d) –7

Expected answer / marking points: (a) 3

Q4 [1]

Question: The slope of the line $y = 5$ is:

(a) 5
(b) 1
(c) 0
(d) not defined

Expected answer / marking points: (c) 0

Q5 [1]

Question: The decimal expansion of an irrational number is:

(a) terminating
(b) non-terminating repeating
(c) non-terminating non-repeating
(d) always zero

Expected answer / marking points: (c) non-terminating non-repeating

Q6 [1]

Question: $\sqrt{5} \times \sqrt{20}$ =

(a) 10
(b) $\sqrt{25}$
(c) 100
(d) $5\sqrt{2}$

Expected answer / marking points: (a) 10

Q7 [1]

Question: Using an identity, 99^2 =

(a) 9801
(b) 9881
(c) 9701
(d) 9999

Expected answer / marking points: (a) 9801

Q8 **[1]**

Question: A factor of $x^2 - 5x + 6$ is:

(a) $x + 2$ (b) $x - 2$

(c) $x + 3$ (d) $x - 6$

Expected answer / marking points: (b) $x - 2$

Q9 **[1]**

Question: A chord equal to the radius subtends an angle at the centre of:

(a) 30° (b) 45°

(c) 60° (d) 90°

Expected answer / marking points: (c) 60°

Q10 **[1]**

Question: The area of a circle of radius 7 cm is:

(a) 44 cm^2 (b) 154 cm^2

(c) 49 cm^2 (d) 22 cm^2

Expected answer / marking points: (b) 154 cm^2

Q11 **[1]**

Question: A bag has 4 red and 6 blue balls. P(red) =

(a) 2/5 (b) 3/5

(c) 1/4 (d) 2/3

Expected answer / marking points: (a) 2/5

Q12 **[1]**

Question: The next term of 1, 4, 9, 16, ... is:

(a) 20 (b) 25

(c) 24 (d) 36

Expected answer / marking points: (b) 25

Q13 **[1]**

Question: The 20th term of the AP 1, 3, 5, ... is:

(a) 39 (b) 40

(c) 41 (d) 38

Expected answer / marking points: (a) 39

Q14 **[1]**

Question: The converse of "If $x = 2$, then $x^2 = 4$" is:

(a) If $x \neq 2$, then $x^2 \neq 4$.

(b) If $x^2 = 4$, then $x = 2$.

(c) If $x^2 \neq 4$, then $x \neq 2$.

(d) $x = 2$ only if $x = 4$.

Expected answer / marking points: (b) If $x^2 = 4$, then $x = 2$. — — false, since x can be –2

Q15 **[1]**

Question: The average of 4 numbers is 25. A fifth number, 35, is added. The new average is:

(a) 26 (b) 27

(c) 30 (d) 28

Expected answer / marking points: (b) 27

Q16 **[1]** **Question:** The number of divisors of 12 is:

(a) 4 (b) 5

(c) 6 (d) 12

Expected answer / marking points: (c) 6

Q17 **[1]** **Question:** The sum of the interior angles of a quadrilateral is:

(a) 180° (b) 270°

(c) 360° (d) 540°

Expected answer / marking points: (c) 360°

Q18 **[1]** **Question:** The total surface area of a cube of edge 4 cm is:

(a) 64 cm^2 (b) 96 cm^2

(c) 16 cm^2 (d) 48 cm^2

Expected answer / marking points: (b) 96 cm^2

Q19 **[1]** **Question: A:** The point (–2, 0) lies on the x-axis.
R: Every point on the x-axis has its y-coordinate equal to 0.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q20 **[1]** **Question: A:** The pair $x + y = 2$ and $2x + 2y = 4$ has infinitely many solutions.
R: If $a_1/a_2 = b_1/b_2 = c_1/c_2$, the lines coincide.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q21 **[2]** **Question:** Write two irrational numbers between 0.1 and 0.2.

Expected answer / marking points: e.g., 0.101001000100001... and 0.1202002000... (non-terminating, non-repeating) [1 + 1].

Q22 **[2]** **Question:** Factorise $6x^2 + 5x – 6$.

Expected answer / marking points: Split $5x = 9x – 4x$: $3x(2x + 3) – 2(2x + 3)$ [1] = $(2x + 3)(3x – 2)$ [1].

Q23 **[2]** **Question:** Find the perimeter of a semicircular protractor of diameter 14 cm.

Expected answer / marking points: $\pi r + d = 22/7 \times 7 + 14$ [1] = 22 + 14 = 36 cm [1].

Q24 [2] **Question:** A coin is tossed 200 times and shows heads 112 times. Find the experimental probability of heads and of tails.

Expected answer / marking points: P(H) = 112/200 = 0.56 [1]; P(T) = 88/200 = 0.44 [1].

Q25 [2] **Question:** Which term of the GP 5, 10, 20, ... is 640?

Expected answer / marking points: $5 \times 2^{n-1} = 640 \Rightarrow 2^{n-1} = 128 = 2^7$ [1] $\Rightarrow n = 8$ [1].

Q26 [3] **Question:** Simplify $(3 + \sqrt{2})(3 - \sqrt{2})$ and $(\sqrt{3} + 1)^2$. State whether each answer is rational or irrational.

Expected answer / marking points: 9 – 2 = 7, rational [1.5]; $3 + 2\sqrt{3} + 1 = 4 + 2\sqrt{3}$, irrational [1.5].

Q27 [3] **Question:** A rhombus has perimeter 40 cm and one diagonal 12 cm. Find the other diagonal and its area.

Expected answer / marking points: Side 10 cm; half-diagonal 6 ⇒ other half = $\sqrt{(100 - 36)} = 8$ [1]; other diagonal 16 cm [1]; area = ½ × 12 × 16 = 96 cm^2 [1].

Q28 [3] **Question:** In a circle of radius 5 cm, AB and AC are two chords of length 6 cm each. Find the distance of each chord from the centre. Which property of chords does this show?

Expected answer / marking points: Distance = $\sqrt{(5^2 - 3^2)} = 4$ cm for each [2]; equal chords are equidistant from the centre [1].

Q29 [3] **Question:** Two dice are thrown together. Find the probability of getting (i) a doublet (ii) a sum of 9 (iii) a product of 6.

Expected answer / marking points: (i) 6/36 = 1/6 [1] (ii) (3,6), (4,5), (5,4), (6,3) ⇒ 4/36 = 1/9 [1] (iii) (1,6), (6,1), (2,3), (3,2) ⇒ 1/9 [1].

Q30 [3] **Question:** Use Euclid's algorithm to find gcd(1071, 462).

Expected answer / marking points: 1071 = 2 × 462 + 147 [1]; 462 = 3 × 147 + 21 [1]; 147 = 7 × 21 + 0 ⇒ gcd = 21 [1].

Q31 [3] **Question:** The cost of 5 kg apples and 3 kg grapes is ₹540, and of 3 kg apples and 4 kg grapes is ₹500. Find the cost of 1 kg of each.

Expected answer / marking points: 5a + 3g = 540, 3a + 4g = 500 [1]; eliminate g: 11a = 660 ⇒ a = 60 [1]; g = 80 [1]. Check 180 + 320 = 500 ✓.

Q32 [5] **Question:** Draw the graphs of 3x + y = 6 and x – y = 2. Find where they intersect, and the area of the triangle formed by the two lines and the y-axis.

ATHVA / OR

(i) If a + b + c = 9 and ab + bc + ca = 26, find $a^2 + b^2 + c^2$. (ii) Factorise $125x^3 - 8y^3$. (iii) Verify $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ for a = 1, b = 2, c = 3.

Expected answer / marking points: Graph [2]; intersection (2, 0) — on the x-axis [1]; y-intercepts (0, 6) and (0, –2) ⇒ base 8 [1]; area = ½ × 8 × 2 = 8 sq units [1].
OR: (i) 81 – 52 = 29 [1.5]. (ii) $(5x - 2y)(25x^2 + 10xy + 4y^2)$ [1.5]. (iii) LHS = 1 + 8 + 27 – 18 = 18; RHS = 6 × (14 – 11) = 18 ✓ [2].

Q33 [5] **Question:** Prove that if the diagonals of a quadrilateral bisect each other, then it is a parallelogram.

ATHVA / OR

ABCD is a trapezium with AB ∥ DC. E is the midpoint of AD. A line through E parallel to AB meets BC at F. Prove that F is the midpoint of BC.

Expected answer / marking points: Figure: ABCD, diagonals meet at O with OA = OC, OB = OD [1]. △AOB ≅ △COD (SAS, vertically opposite angles) [1.5] ⇒ ∠OAB = ∠OCD ⇒ AB ∥ CD (alternate angles) [1]. Similarly △AOD ≅ △COB ⇒ AD ∥ BC [1]. Both pairs of opposite sides parallel ⇒ parallelogram [0.5].
OR: Join AC, meeting EF at G [1]. In △ADC, E is the midpoint of AD and EG ∥ DC ⇒ G is the midpoint of AC (converse of Midpoint Theorem) [2]. In △CAB, G is the midpoint of AC and GF ∥ AB ⇒ F is the midpoint of BC [2].

Q34 [5] **Question:** A park is in the shape of quadrilateral ABCD with ∠C = 90°, AB = 9 m, BC = 12 m, CD = 5 m and AD = 8 m. Find its area.

Expected answer / marking points: BD = √(144 + 25) = 13 m [1]; △BCD = ½ × 12 × 5 = 30 m^2 [1]; △ABD: s = 15, √(15 × 6 × 7 × 2) = √1260 ≈ 35.5 m^2 [2]; total ≈ 65.5 m^2 [1].

Q35 [5] **Question:** A hemispherical bowl has an inner radius of 10.5 cm. (i) How much water can it hold, in litres? (ii) Find the cost of tin-plating its inner surface at ₹16 per 100 cm^2. (iii) How many cylindrical cups of radius 3.5 cm and height 9 cm can be filled from the full bowl?

Expected answer / marking points: (i) ⅔ × 22/7 × 10.5^3 = 2425.5 cm^3 ≈ 2.43 L [2]. (ii) $2\pi r^2$ = 693 cm^2; cost = ₹110.88 [1.5]. (iii) One cup = 22/7 × 12.25 × 9 = 346.5 cm^3; 2425.5 ÷ 346.5 = 7 cups [1.5].

Q36 [4] **Question:**

> In a quiz league, Team A played 4 rounds with an average score of 15. Team B played 6 rounds with an average score of 20.

(i) Find Team A's total score. [1]
(ii) Find the average score of all 10 rounds together. [1]
(iii) Team A plays one more round. What must it score to raise its average to 16? [2]

Expected answer / marking points: (i) 60 [1]. (ii) (60 + 120)/10 = 18 [1]. (iii) (60 + x)/5 = 16 [1] ⇒ x = 20 [1].

Q37 [4] **Question:**

> A, B, C and D are four points on a circular running track with centre O.

(i) If ∠ACB = 35°, find ∠AOB. [1]
(ii) If AB is a diameter, what is ∠ACB? [1]
(iii) If ABCD is a cyclic quadrilateral with ∠ABC = 80° and ∠BAD = 95°, find ∠ADC and ∠BCD. [2]

Expected answer / marking points: (i) 70° [1]. (ii) 90° [1]. (iii) ∠ADC = 180° – 80° = 100° [1]; ∠BCD = 180° – 95° = 85° [1].

Q38 **Question:**

[4]

> In a Sierpiński carpet, a square is divided into 9 equal squares and the middle one is removed. At every stage, each remaining square is treated the same way.

(i) How many small squares remain after stage 2? [1]
(ii) Write the number remaining after stage n. What type of progression is it? [1]
(iii) If the original square has area 81 cm^2, find the remaining area after stage 2. [2]

Expected answer / marking points: (i) $8 \times 8 = 64$ [1]. (ii) 8^n; GP with r = 8 [1]. (iii) Each stage keeps 8/9 of the area: $81 \times (8/9)^2$ [1] = 64 cm^2 [1].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Science · Mock Paper 1 · Answer key

Wednesday, 17 February 2027 · 80 marks

Test ID: T2027-02-17

Class 9 · Exploration — poora syllabus (Ch 1–13)

Q1 [1]

Question: A fair test in an experiment means:

(a) changing many variables at once

(b) changing only one variable and keeping others the same

(c) not recording results

(d) repeating only once

Expected answer / marking points: (b) changing only one variable and keeping others the same

Q2 [1]

Question: Which organelle is found in plant cells but not in animal cells?

(a) Mitochondria (b) Chloroplast

(c) Ribosome (d) Nucleus

Expected answer / marking points: (b) Chloroplast

Q3 [1]

Question: The tissue that helps in the growth of plants at the tips of stems and roots is:

(a) parenchyma (b) apical meristem

(c) xylem (d) cork

Expected answer / marking points: (b) apical meristem

Q4 [1]

Question: An object moves 20 m in 4 s and another 30 m in the next 6 s. Its average speed is:

(a) 5 m s^{-1} (b) 50 m s^{-1}

(c) 10 m s^{-1} (d) 2.5 m s^{-1}

Expected answer / marking points: (a) 5 m s^{-1}

Q5 [1]

Question: A mixture of two miscible liquids with a large difference in boiling points is separated by:

(a) filtration (b) distillation

(c) sublimation (d) chromatography

Expected answer / marking points: (b) distillation

Q6 [1]

Question: The SI unit of momentum is:

(a) kg m s^{-1} (b) N

(c) kg m s^{-2} (d) J

Expected answer / marking points: (a) kg m s^{-1}

Q7 [1]

Question: 1 kWh is equal to:

(a) 1000 J (b) 3600 J

(c) 3.6×10^6 J (d) 36 J

Expected answer / marking points: (c) 3.6×10^6 J

Q8 [1]

Question: Rutherford's gold foil experiment showed that:

(a) electrons are in fixed orbits

(b) the atom has a small, dense, positive nucleus

(c) neutrons exist

(d) atoms are indivisible

Expected answer / marking points: (b) the atom has a small, dense, positive nucleus

Q9 [1]

Question: The number of electrons in the outermost shell of sodium (Z = 11) is:

(a) 1 (b) 2

(c) 8 (d) 11

Expected answer / marking points: (a) 1

Q10 [1]

Question: The formula of aluminium chloride is:

(a) AlCl (b) $AlCl_2$

(c) $AlCl_3$ (d) Al_3Cl

Expected answer / marking points: (c) $AlCl_3$

Q11 [1]

Question: The speed of a sound wave of frequency 680 Hz and wavelength 0.5 m is:

(a) 340 m s^{-1} (b) 1360 m s^{-1}

(c) 680 m s^{-1} (d) 170 m s^{-1}

Expected answer / marking points: (a) 340 m s^{-1}

Q12 [1]

Question: Bats find their way using:

(a) infrasound (b) ultrasound (echolocation)

(c) light (d) smell only

Expected answer / marking points: (b) ultrasound (echolocation)

Q13 [1]

Question: Yeast reproduces asexually mainly by:

(a) binary fission (b) budding

(c) spore formation only (d) fragmentation

Expected answer / marking points: (b) budding

Q14 [1]

Question: The correct order of classification levels is:

(a) Kingdom → Class → Phylum → Order

(b) Kingdom → Phylum → Class → Order

(c) Class → Kingdom → Order → Phylum

(d) Order → Class → Phylum → Kingdom

Expected answer / marking points: (b) Kingdom → Phylum → Class → Order

Q15 [1] **Question:** Most of the weather (clouds, rain, storms) occurs in the:

(a) stratosphere (b) troposphere

(c) mesosphere (d) thermosphere

Expected answer / marking points: (b) troposphere

Q16 [1] **Question:** Nitrogen fixation is mainly done by:

(a) fungi (b) bacteria (and lightning)

(c) animals (d) volcanoes

Expected answer / marking points: (b) bacteria (and lightning)

Q17 [1] **Question: A:** Iron filings can be separated from sulphur powder using a magnet.
R: Iron is attracted by a magnet but sulphur is not.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q18 [1] **Question: A:** A body moving with uniform velocity has zero acceleration.
R: Acceleration is the rate of change of velocity.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q19 [1] **Question: A:** Echo is heard only when the reflecting surface is at least 17 m away (in air).
R: The sensation of sound persists in our brain for about 0.1 s.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q20 [1] **Question: A:** The binomial name of the mango is Mangifera indica.
R: In binomial nomenclature, both genus and species names start with capital letters.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (c) — A true; R false (species starts with a small letter).

Q21 **Question:** What is the function of (i) xylem (ii) phloem?
[2]

Expected answer / marking points: (i) Carries water and minerals from roots upward [1] (ii) Carries food from leaves to other parts [1].

Q22 **Question:** A cyclist moving at 5 m s^{-1} speeds up to 15 m s^{-1} in 4 s. Find the acceleration.
[2]

Expected answer / marking points: a = (15 – 5)/4 [1] = 2.5 m s^{-2} [1].

Q23 **Question:** What is chromatography? Give one use.
[2]

Expected answer / marking points: A method to separate the components of a mixture that dissolve in the same solvent, based on how fast they move [1]; to separate dyes in ink/pigments, or detect drugs in blood [1].

Q24 **Question:** Why does a fielder pull his hands back while catching a fast cricket ball?
[2]

Expected answer / marking points: It increases the time to stop the ball [1]; so the force (rate of change of momentum) on the hands is less [1].

Q25 **Question:** Write two differences between sexual and asexual reproduction.
[2]

Expected answer / marking points: Sexual: two parents, gametes fuse; asexual: one parent, no gametes [1]; sexual gives variation, asexual gives identical offspring [1].

Q26 **Question:** Give two effects of climate change on the water cycle.
[2]

Expected answer / marking points: More evaporation → heavier rain and floods in some places [1]; droughts, melting glaciers, changed monsoon patterns (any one) [1].

Q27 **Question:** Distinguish between the three types of muscles on the basis of their structure and control.
[3]

Expected answer / marking points: Skeletal: long, striated, multinucleate, voluntary [1]; smooth: spindle-shaped, unstriated, involuntary [1]; cardiac: branched, striated, involuntary, never tire [1].

Q28 **Question:** A train travelling at 72 km h^{-1} applies brakes and stops in 20 s. Find its retardation and the distance it covers before stopping.
[3]

Expected answer / marking points: u = 20 m s^{-1} [0.5]; a = –20/20 = –1 m s^{-2} [1]; s = 20 × 20 – ½ × 1 × 400 = 200 m [1.5].

Q29 **Question:** How would you separate a mixture of salt, sand and camphor? Write the steps in order.
[3]

Expected answer / marking points: Heat — camphor sublimes; collect it on a cool surface [1]; add water to the rest, salt dissolves; filter to remove sand [1]; evaporate the filtrate to get salt [1].

Q30 **Question:** A 10 kg box is pushed with a force of 50 N against a friction force of 20 N on a level floor. Find (i) the net force (ii) the acceleration (iii) the velocity after 4 s if it starts from rest.
[3]

Expected answer / marking points: (i) 30 N [1] (ii) 3 m s^{-2} [1] (iii) 12 m s^{-1} [1].

Q31 **Question:** Write the electronic configurations of carbon (6), oxygen (8) and chlorine (17) and their valencies.
[3]

Expected answer / marking points: C: 2, 4 — valency 4 [1]; O: 2, 6 — valency 2 [1]; Cl: 2, 8, 7 — valency 1 [1].

Q32 [3] **Question:** Draw a sound wave as a density–distance graph and mark wavelength and amplitude. How is loudness related to amplitude?

Expected answer / marking points: Correct graph with compressions/rarefactions [1]; wavelength and amplitude marked [1]; greater amplitude ⇒ louder sound [1].

Q33 [3] **Question:** Describe vegetative propagation by stem, root and leaf with one example each.

Expected answer / marking points: Stem — rose/sugarcane cuttings, potato (tuber) [1]; root — sweet potato, dahlia [1]; leaf — Bryophyllum (buds on leaf margin) [1].

Q34 [5] **Question:** Derive $v = u + at$ and $s = ut + \frac{1}{2}at^2$ using a velocity–time graph. A car moving at 10 m s^{-1} accelerates at 2 m s^{-2} for 5 s. Find its final velocity and distance covered.

ATHVA / OR

State the law of conservation of momentum. A 50 g bullet is fired from a 2 kg pistol at 150 m s^{-1}. Find the recoil velocity. Explain why a rocket moves upward using the third law.

Expected answer / marking points: v–t graph derivations [3]; v = 10 + 10 = 20 m s^{-1} [1]; s = 50 + 25 = 75 m [1].
OR: Statement [1]; 0.05 × 150 + 2v = 0 ⇒ v = –3.75 m s^{-1} [2]; rocket pushes hot gases down (action), gases push the rocket up (reaction) [2].

Q35 [5] **Question:** Describe the models of the atom by Thomson, Rutherford and Bohr, giving one drawback or improvement of each.

ATHVA / OR

Explain the five-kingdom system of classification. Why are fungi kept in a separate kingdom from plants?

Expected answer / marking points: Thomson — positive sphere with embedded electrons (plum pudding); could not explain gold foil results [1.5]; Rutherford — tiny dense nucleus, electrons revolve; could not explain stability [1.5]; Bohr — electrons in fixed energy orbits without losing energy; explained stability [2].
OR: Five kingdoms with key features + examples [3.5]; fungi are heterotrophic (no chlorophyll), cell wall of chitin, absorb food from dead matter [1.5].

Q36 [5] **Question:** Explain the role of uneven heating of the Earth in causing (i) sea and land breezes (ii) planetary winds (iii) ocean currents. How does albedo affect a region's temperature?

Expected answer / marking points: (i) Day: land heats faster → sea breeze; night: land cools faster → land breeze [1.5]; (ii) equator hotter → air rises, cooler air moves in → planetary winds deflected by Earth's rotation [1]; (iii) winds and temperature differences drive currents carrying heat [1]; high albedo (ice) reflects more → cooler; low albedo (forest, ocean) absorbs → warmer [1.5].

Q37 **Question:**

[4]

> A 40 kg student uses a ramp (inclined plane) 6 m long to push a 20 kg box onto a stage 1.5 m high instead of lifting it straight up.

(i) Find the work done against gravity in raising the box. [1]
(ii) If she applies a force of 60 N along the ramp, what work does she do? [1]
(iii) Why is the work done along the ramp more than the work against gravity? How does the ramp help her? [2]

Expected answer / marking points: (i) mgh = 20 × 10 × 1.5 = 300 J [1]. (ii) 60 × 6 = 360 J [1]. (iii) Extra work is used against friction [1]; she needs a smaller force (60 N instead of 200 N) over a longer distance [1].

Q38 **Question:**

[4]

> Element P has electronic configuration 2, 8, 2 and element Q has 2, 8, 7.

(i) Identify P and Q. [1]
(ii) What type of bond will they form? [1]
(iii) Write the formula of the compound and two properties you expect it to have. [2]

Expected answer / marking points: (i) P = Mg, Q = Cl [1]. (ii) Ionic [1]. (iii) $MgCl_2$ [1]; high melting point, soluble in water, conducts in solution/molten state (any two) [1].

Q39 **Question:**

[4]

> The Western Ghats and the Eastern Himalayas are among the world's biodiversity hotspots. Many species there are found nowhere else, but forests are being cleared for roads and plantations.

(i) What is a biodiversity hotspot? [1]
(ii) What are endemic species? [1]
(iii) Give two threats to biodiversity mentioned or implied and one way to protect these regions. [2]

Expected answer / marking points: (i) A region with very high biodiversity that is under threat [1]. (ii) Species found only in a particular region [1]. (iii) Habitat loss/deforestation, roads/plantations (also poaching, pollution) [1]; protected areas/national parks, laws, community conservation [1].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Social Science · Mock Paper 1 · Answer key

Thursday, 18 February 2027 · 80 marks

Test ID: T2027-02-18

Class 9 · Part 1 (Ch 1–9) + Part 2 (Ch 1–6) — poora syllabus

Q1 **[1]**

Question: Earthquakes and volcanoes are caused by:

(a) external forces
(b) internal (endogenic) forces
(c) wind
(d) glaciers

Expected answer / marking points: (b) internal (endogenic) forces

Q2 **[1]**

Question: A V-shaped valley is formed by:

(a) a glacier
(b) a river in its upper course
(c) wind
(d) sea waves

Expected answer / marking points: (b) a river in its upper course

Q3 **[1]**

Question: The ozone layer is found in the:

(a) troposphere
(b) stratosphere
(c) mesosphere
(d) exosphere

Expected answer / marking points: (b) stratosphere

Q4 **[1]**

Question: India's climate is described as:

(a) Mediterranean
(b) tropical monsoon
(c) polar
(d) temperate

Expected answer / marking points: (b) tropical monsoon

Q5 **[1]**

Question: The period when farming and domestication of animals began is the:

(a) Palaeolithic
(b) Mesolithic
(c) Neolithic
(d) Bronze Age

Expected answer / marking points: (c) Neolithic

Q6 **[1]**

Question: Magadha, the Mauryas and the Guptas are associated with:

(a) early states and empires in India
(b) European kingdoms
(c) Mughal rulers
(d) British rule

Expected answer / marking points: (a) early states and empires in India

Q7 **[1]**

Question: The minimum age for voting in India is:

(a) 16
(b) 18
(c) 21
(d) 25

Expected answer / marking points: (b) 18

Q8 [1]

Question: Which of the following is NOT a function of the Election Commission?

(a) Preparing voter lists
(b) Allotting election symbols
(c) Making laws for the country
(d) Enforcing the Model Code of Conduct

Expected answer / marking points: (c) Making laws for the country

Q9 [1]

Question: Opportunity cost is:

(a) the price of a good
(b) the value of the next best alternative given up
(c) the total cost of production
(d) tax paid

Expected answer / marking points: (b) the value of the next best alternative given up

Q10 [1]

Question: If the price of petrol rises sharply, the demand for public transport will most likely:

(a) fall
(b) rise
(c) stay the same
(d) become zero

Expected answer / marking points: (b) rise

Q11 [1]

Question: Deep, narrow depressions on the ocean floor are called:

(a) ridges
(b) trenches
(c) shelves
(d) seamounts

Expected answer / marking points: (b) trenches

Q12 [1]

Question: Tides are mainly caused by the gravitational pull of the:

(a) Sun only
(b) Moon (and the Sun)
(c) Mars
(d) winds

Expected answer / marking points: (b) Moon (and the Sun)

Q13 [1]

Question: The Chipko movement is linked with protecting:

(a) rivers
(b) forests/trees
(c) wildlife only
(d) mountains

Expected answer / marking points: (b) forests/trees

Q14 [1]

Question: Maharana Pratap fought the Mughals at the Battle of Haldighati in:

(a) 1526
(b) 1576
(c) 1671
(d) 1707

Expected answer / marking points: (b) 1576

Q15 [1]

Question: Chhatrapati Shivaji was crowned in:

(a) 1659
(b) 1674
(c) 1680
(d) 1707

Expected answer / marking points: (b) 1674

Q16 [1] **Question:** Public policy means:

(a) decisions of private companies

(b) government decisions and actions to solve public problems

(c) laws made by courts

(d) opinions of the media

Expected answer / marking points: (b) government decisions and actions to solve public problems

Q17 [1] **Question:** Insurance helps a person by:

(a) increasing income

(b) protecting against financial loss

(c) reducing taxes to zero

(d) giving loans

Expected answer / marking points: (b) protecting against financial loss

Q18 [1] **Question:** Keeping money idle for many years is risky mainly because of:

(a) interest

(b) inflation

(c) profit

(d) budgeting

Expected answer / marking points: (b) inflation

Q19 [1] **Question:** Turning an idea into action by taking risks and creating value is called:

(a) entrepreneurship

(b) employment

(c) taxation

(d) saving

Expected answer / marking points: (a) entrepreneurship

Q20 [1] **Question:** Viksit Bharat is a vision for the year:

(a) 2030

(b) 2040

(c) 2047

(d) 2050

Expected answer / marking points: (c) 2047

Q21 [2] **Question:** What is a delta? How is it formed?

Expected answer / marking points: A triangular landform at a river's mouth [1]; formed when a river slows down and deposits sediment before entering the sea [1].

Q22 [2] **Question:** What were the sabhā and samiti? How did they limit the power of the rājā?

Expected answer / marking points: Vedic assemblies [1]; the rājā had to consult them, so decisions were collective and he could not rule arbitrarily [1].

Q23 [2] **Question:** Give two differences between national and state (regional) parties.

Expected answer / marking points: National parties are recognised in several states / contest across India; state parties mainly in one state [1]; focus on all-India vs regional issues [1].

Q24 [2] **Question:** What is the Exclusive Economic Zone? Why is it important for India?

Expected answer / marking points: Sea area up to 200 nautical miles where a country has exclusive rights over resources [1]; gives India fish, oil, gas and minerals [1].

Q25 **Question:** Distinguish between the south-west monsoon and the north-east monsoon.
[3]

Expected answer / marking points: SW monsoon: June–September, from the Arabian Sea and Bay of Bengal, gives most of India's rain [1.5]; NE monsoon: October–December, gives rain mainly to Tamil Nadu coast [1.5].

Q26 **Question:** What can we learn from early Indian society about education and the position of women? Give three points.
[3]

Expected answer / marking points: Education emphasised knowledge and moral values (gurukulas, centres like Takshashila/Nalanda) [1]; some women were learned and took part in debates [1]; position of women varied over time and region [1].

Q27 **Question:** Explain three reforms by the Election Commission to make voting more inclusive.
[3]

Expected answer / marking points: Home voting for senior citizens (85+) and persons with disabilities [1]; postal ballots for service voters [1]; accessible booths, ramps, braille, volunteers (any) [1].

Q28 **Question:** Why do pure market or pure planned economies rarely exist? Why is a mixed economy considered practical?
[3]

Expected answer / marking points: Pure market may ignore the poor and public goods; pure planned may limit freedom and innovation [1.5]; mixed economy combines private initiative with government welfare/regulation [1.5].

Q29 **Question:** How did economic strength help communities stay resilient between 1000 and 1700 CE?
[3]

Expected answer / marking points: Agriculture and irrigation gave food security [1]; markets, merchant networks and trade gave resources [1]; financial practices and guilds helped communities recover from instability [1].

Q30 **Question:** Explain the composition and structure of the atmosphere with a labelled diagram of its layers.
[5]

ATHVA / OR

What is climate change? Explain its main causes and three ways in which we can reduce our carbon footprint.

Expected answer / marking points: Composition — nitrogen ~78%, oxygen ~21%, argon, CO_2, water vapour, dust [2]; layers — troposphere (weather), stratosphere (ozone), mesosphere, thermosphere, exosphere with one feature each [2]; diagram [1].
OR: Long-term change in weather patterns [1]; causes — burning fossil fuels, deforestation, industries, vehicles [2]; reduce — save energy, renewable energy, public transport, plant trees, reduce waste [2].

Q31 **Question:** Describe the main steps in the conduct of elections in India, from delimitation to the declaration of results.
[5]

Expected answer / marking points: Delimitation of constituencies; voter registration/electoral rolls; announcement and Model Code; nominations and scrutiny; campaigning; polling (EVM/VVPAT); counting and declaration [5 × 1, any five in order].

Q32 [5] **Question:** Explain any five factors that affect the demand for a commodity, with examples.

ATHVA / OR

What is a business plan? Explain the four functional areas of a business with an example of a bakery.

Expected answer / marking points: Price of the good; income; prices of substitutes/complements; tastes and preferences; season; expectations; population — any five with examples [1 × 5].
OR: Business plan — roadmap of goals, resources, decisions [1]; management & operations (baking schedule), finance & accounting (costs, profit), human resources (hiring bakers), marketing & sales (ads, pricing) [4].

Q33 [5] **Question:** Explain five principles of good governance and show how any one government scheme reflects them.

Expected answer / marking points: Participation, transparency, accountability, rule of law, responsiveness/efficiency explained [3.5]; example scheme (e.g., Swachh Bharat, Digital India) linked to principles [1.5].

Q34 [4] **Question:**

> Ravi's family earns ₹40,000 a month. They spend ₹28,000 on household needs, ₹4,000 on EMIs, save ₹5,000 in a recurring deposit, and pay ₹3,000 as premiums for health and life insurance.

(i) What percentage of income does the family save? [1]
(ii) Which pillar of personal finance do the premiums represent? [1]
(iii) Suggest two ways the family can improve its finances for long-term goals. [2]

Expected answer / marking points: (i) 5000/40000 × 100 = 12.5% [1]. (ii) Protection [1]. (iii) Make a budget and cut unnecessary spending; invest part of savings in options that beat inflation; build an emergency fund (any two) [2].

Q35 [4] **Question:**

> A sudden heavy rain at the start of the season damaged a large part of the tomato crop in a district. Within a week, the price of tomatoes in the city doubled. Some people started buying less, and a few switched to other vegetables.

(i) What happened to the supply of tomatoes? [1]
(ii) Why did the price rise? [1]
(iii) Explain the change in quantity demanded using the law of demand, and suggest one step the government could take. [2]

Expected answer / marking points: (i) Supply fell [1]. (ii) Less supply with the same demand ⇒ shortage ⇒ price rises [1]. (iii) Higher price ⇒ lower quantity demanded; people shift to substitutes [1]; government could release stocks, improve storage, or arrange supply from other regions [1].

Q36 [4] **Question:**

> In 1671, the Mughal army tried to take control of the Brahmaputra valley. The Ahom commander Lachit Borphukan used the narrow river at Saraighat, boats and the local terrain to defeat a much larger Mughal fleet.

(i) Which dynasty did Lachit Borphukan serve? [1]
(ii) How did geography help the Ahoms? [1]
(iii) Is this an example of resistance or resilience? Explain, and give one more example from the chapter. [2]

Expected answer / marking points: (i) The Ahoms [1]. (ii) Narrow river and local knowledge of terrain helped a smaller force defeat a larger fleet [1]. (iii) Resistance — actively opposing a threat to protect territory [1]; e.g., Maharana Pratap at Haldighati / Shivaji / Kumara Kampana [1].

Q37 [5] **Question:** On an outline political map of India, locate and label any five: (a) the Bay of Bengal (b) the Western Ghats (c) the Brahmaputra river (d) Saraighat/Guwahati (e) Panipat (f) Madurai (g) the Andaman and Nicobar Islands.

Expected answer / marking points: Each correctly located and labelled [1 × 5].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

English · Mock Paper 1 · Answer key

Friday, 19 February 2027 · 80 marks

Test ID: T2027-02-19

Class 9 · Reading · Grammar · Writing · Literature (Kaveri Units 1–8)

Q1 [10] **Question:** Read the passage and answer the questions.

> 1. The Indian Railways runs more than 13,000 passenger trains every day and carries over two crore passengers — roughly the population of a large country. Behind this daily movement is a network of about 68,000 km of track, over 7,000 stations and more than twelve lakh employees.
> 2. Trains have been running in India since 1853, when the first passenger train travelled 34 km from Bori Bunder in Bombay to Thane. Since then, steam engines have given way to diesel and electric locomotives. Today, most of the main routes are electrified, which reduces pollution and saves fuel.
> 3. The railways also carry coal, food grains, cement and steel across the country. Moving goods by rail uses much less fuel per tonne than moving them by road, so it is better for the environment.
> 4. New semi-high-speed trains have cut travel time on several routes. Stations are being redeveloped with ramps, lifts and clean waiting areas so that elderly passengers and people with disabilities can travel with dignity.
> 5. Yet the railways face challenges: overcrowded general coaches, safety at unmanned crossings and the need to maintain old bridges. Passengers too have a role — keeping trains clean and respecting fellow travellers.

(i) About how many passengers do Indian trains carry every day? [1]
(ii) When and between which places did the first passenger train run? [1]
(iii) Name two types of locomotives that replaced steam engines. [1]
(iv) Give one benefit of electrification. [1]
(v) Name two goods carried by rail. [1]
(vi) Why is moving goods by rail better for the environment? [1]
(vii) How are stations being made easier for elderly passengers? [1]
(viii) State one challenge the railways face. [1]
(ix) Find a word in paragraph 5 that means “too full of people”. [1]
(x) What role do passengers have, according to the passage? [1]

Expected answer / marking points: (i) Over two crore. (ii) 1853, Bori Bunder (Bombay) to Thane. (iii) Diesel and electric. (iv) Less pollution / saves fuel. (v) Any two: coal, food grains, cement, steel. (vi) Uses less fuel per tonne than road. (vii) Ramps, lifts, clean waiting areas. (viii) Overcrowding / safety at unmanned crossings / old bridges. (ix) overcrowded. (x) Keep trains clean and respect fellow travellers. [1 × 10]

Q2 [10] **Question:** Read the passage and answer the questions.

> 1. When thirteen-year-old Tenzin moved from a small town in Ladakh to Delhi, the noise and speed of the city frightened her. In class, she understood the lessons but was too shy to speak. Her Hindi had a Ladakhi accent, and a few students laughed when she answered once.
> 2. Her English teacher noticed that Tenzin wrote beautiful essays. She asked Tenzin to write about her home for the school magazine. Tenzin wrote about apricot orchards, frozen rivers in winter and her grandmother's butter tea.
> 3. The article surprised everyone. Classmates came up to ask about Ladakh — whether it really snowed in April, and how cold minus twenty felt. One of the boys who had laughed apologised and asked her to teach him a few Ladakhi words.
> 4. By the end of the year, Tenzin was leading the school's "Know Your India" club, where students shared stories from their home states. "Being different is not a weakness," she says. "It is something new you can give to others."

(i) Why did Tenzin find Delhi frightening at first? [1]
(ii) Why was she too shy to speak in class? [2]
(iii) What did the English teacher notice and what did she suggest? [2]
(iv) Name two things Tenzin wrote about. [1]
(v) How did one of the boys show a change in his behaviour? [1]
(vi) What club did Tenzin lead? [1]
(vii) Find a word in paragraph 3 that means "said sorry". [1]
(viii) What does Tenzin mean by "Being different is not a weakness"? [1]

Expected answer / marking points: (i) The noise and speed of the city [1]. (ii) Her Hindi had a Ladakhi accent and some students laughed when she answered [2]. (iii) That she wrote beautiful essays; asked her to write about her home for the magazine [2]. (iv) Apricot orchards, frozen rivers, grandmother's butter tea (any two) [1]. (v) He apologised and asked her to teach him Ladakhi words [1]. (vi) "Know Your India" club [1]. (vii) apologised [1]. (viii) Our differences give us something new to share with others [1].

Q3 [10] **Question:** Do as directed (1 mark each):
(i) Fill in with a / an / the: Khetaram is ___ only postman of Somarad.
(ii) Fill in with the correct form: Shruti ___ (practise) the violin since she was six.
(iii) Fill in with a modal: ___ I borrow your pen, please?
(iv) Change into passive: The council summoned Mesoba.
(v) Change into reported speech: Avva said, "I want to be independent."
(vi) Correct the error: Grandpa and Ravi was playing chess.
(vii) Combine using "who": Deepa Malik is a para-athlete. She won a silver medal at Rio.
(viii) Fill in with the correct determiner: There are ___ apples in the basket — only two.
(ix) Change into reported speech: The teacher said to us, "Don't waste time."
(x) Fill in with the correct preposition: Ming's mother wrote a letter ___ her daughter.

Expected answer / marking points: (i) the (ii) has been practising / has practised (iii) May/Can (iv) Mesoba was summoned by the council. (v) Avva said that she wanted to be independent. (vi) Grandpa and Ravi were playing chess. (vii) Deepa Malik is a para-athlete who won a silver medal at Rio. (viii) few / only a few (ix) The teacher told/advised us not to waste time. (x) to [1 × 10]

Q4 [5] **Question:** Write an article (100–120 words) for your school magazine on "Why Every Student Should Learn a Craft".

ATHVA / OR

Write a story (100–120 words) beginning with: "The bus had just left when Meera realised her bag was still on the seat..."

Expected answer / marking points: Title + byline [0.5]; introduction, reasons (patience, creativity, respect for artisans, livelihood), conclusion [3]; language and organisation [1.5].
OR: Plot and development from the given beginning [2.5]; characters/dialogue and ending [1]; language, creativity [1.5].

Q5 [5] **Question:** Write a formal email (100–120 words) to the Station Master of your city railway station suggesting two improvements for elderly passengers.

ATHVA / OR

Write a diary entry (100–120 words) about the day your class organised a sports event for children with special needs.

Expected answer / marking points: Format [1.5]; content — two clear suggestions with reasons [2.5]; polite tone/language [1].
OR: Diary format [1]; events, your role, feelings, lesson learnt [2.5]; language [1.5].

Q6 [5] **Question:** Read the stanza and answer.

> If words could satisfy the heart,
> The heart might find less care;
> But words, like summer birds, depart,
> And leave but empty air.

(i) To what are words compared in line 3? [1]
(ii) Name the figure of speech used in "words, like summer birds". [1]
(iii) What do words leave behind, according to the poet? [1]
(iv) Write the rhyme scheme of the stanza. [1]
(v) What does the poet mean by "find less care"? [1]

Expected answer / marking points: (i) Summer birds [1]. (ii) Simile [1]. (iii) Empty air [1]. (iv) abab [1]. (v) The heart would have fewer worries/troubles [1].

Q7 **[5]**

Question: Read the stanza and answer.

> I cannot remember my mother
> only when from my bedroom window I send
> my eyes into the blue of the distant sky,
> I feel that the stillness of
> my mother's gaze on my face
> has spread all over the sky.

(i) Where is the poet when he thinks of his mother here? [1]
(ii) What does he look at? [1]
(iii) Which sense is used in this stanza? [1]
(iv) What does the poet feel has spread all over the sky? [1]
(v) What does this show about the poet's feelings? [1]

Expected answer / marking points: (i) In his bedroom, at the window [1]. (ii) The blue of the distant sky [1]. (iii) Sight (visual) [1]. (iv) The stillness of his mother's gaze [1]. (v) Deep love and longing; he feels her presence everywhere [1].

Q8 **[2]**

Question: What did the narrator in "How I Taught My Grandmother to Read" later become? How was Avva her "first student"? (Unit 1)

Expected answer / marking points: She became a teacher of Computer Science [1]; she taught Avva the Kannada alphabet before teaching anyone else [1].

Q9 **[2]**

Question: Why was Arenla against pot making as a career for Sentila? Give two reasons. (Unit 2)

Expected answer / marking points: Hard labour — clay carried 16 km uphill, tiring pounding [1]; very low earnings (a pittance) compared to weaving [1].

Q10 **[2]**

Question: How did pankhas change from personal objects to commercial goods? (Unit 3)

Expected answer / marking points: Once made for personal/temple/royal use [1]; now crafted for sale and decoration, giving artisans a livelihood [1].

Q11 **[2]**

Question: What did Grandpa do on his day out that he was not allowed at home? (Unit 4)

Expected answer / marking points: Had sugary tea, ate bananas and ice cream (banned items) [1]; went out alone and took a bus [1].

Q12 **[2]**

Question: What does the title "Twin Melodies" suggest? (Unit 6)

Expected answer / marking points: Two kinds of music — classical and Western/fusion — coming together [1]; also harmony between father and daughter [1].

Q13 **[2]**

Question: Why does Khetaram sometimes postpone deliveries? (Unit 7)

Expected answer / marking points: Severe summer heat (around 50 °C) [1]; he waits till after sundown to deliver [1].

Q14 [6] **Question:** Describe Deepa Malik's journey from tragedy to triumph. What qualities helped her? (120–150 words) (Unit 5)

ATHVA / OR

How does the poem "Nine Gold Medals" redefine winning? (120–150 words) (Unit 5)

Expected answer / marking points: Spinal tumour at 29, paralysed below the waist [1]; chose sports, switched to para-athletics; silver in shot put at Rio 2016; many "firsts" [2]; qualities — resilience, fortitude, positive attitude, family support [2]; expression [1].
OR: Story of the race and the fall [1.5]; runners return to help, finish together [1.5]; winning = kindness, unity, humanity over medals [2]; expression [1].

Q15 [6] **Question:** How does the conflict between Shruti and her father get resolved in "Twin Melodies"? What does this teach about communication between parents and children? (120–150 words)

ATHVA / OR

The postal service is described as a symbol of trust in "Carrier of Words". Explain with examples. (120–150 words)

Expected answer / marking points: Conflict — Nabin's dislike of fusion, Shruti hides practice [1.5]; resolution — parents watch practice, father appreciates raga-based fusion, offers music room [2]; lesson — honest talk, openness, respect for both tradition and new ideas [1.5]; expression [1].
OR: Villagers rely on Khetaram to read/write letters [1.5]; savings and money orders through post office [1.5]; GDS scheme kept because villagers trusted them [1.5]; expression [1.5].

Q16 [6] **Question:** "Dreams take effort, sacrifice and time." Discuss with reference to the letter in "Follow That Dream" and the poem in Unit 8. (120–150 words)

Expected answer / marking points: Letter — ten years of focus, counting the cost, support network, changing dreams [2.5]; poem — leaving comfort, first step hardest, self-belief [2]; expression [1.5].

Score: ___ / 80 Checked by: _____

Weak topic aur correction:

Artificial Intelligence (417) · Mock Paper · Answer key

Saturday, 20 February 2027 · 50 marks

Test ID: T2027-02-20

Class 9 · Part A: Employability Skills · Part B: AI Reflection & Project Cycle, Data Literacy, Math for AI, Generative AI, Python

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Q1 [1]

Question: Which of these is an example of active listening?

(a) Interrupting the speaker
(b) Nodding and asking questions
(c) Looking at your phone
(d) Thinking of your reply only

Expected answer / marking points: (b) Nodding and asking questions

Q2 [1]

Question: The "T" in SMART goals stands for:

(a) Tough
(b) Time-bound
(c) Targeted
(d) Tested

Expected answer / marking points: (b) Time-bound

Q3 [1]

Question: Which shortcut key is used to save a file?

(a) Ctrl + S
(b) Ctrl + P
(c) Ctrl + Z
(d) Ctrl + X

Expected answer / marking points: (a) Ctrl + S

Q4 [1]

Question: A person who starts a business by taking risks is called:

(a) an employee
(b) an entrepreneur
(c) a customer
(d) a supplier

Expected answer / marking points: (b) an entrepreneur

Q5 [1]

Question: Which of these is a green job?

(a) Solar panel technician
(b) Coal miner
(c) Plastic bag maker
(d) Diesel generator seller

Expected answer / marking points: (a) Solar panel technician

Q6 [1]

Question: Which of these is a renewable energy source?

(a) Coal
(b) Petrol
(c) Wind
(d) Natural gas

Expected answer / marking points: (c) Wind

Q7 [2]

Question: Write two ways to manage stress during exams.

Expected answer / marking points: Any two: proper sleep, exercise/play, timetable with breaks, talking to family/friends, deep breathing [2].

Q8 **Question:** Differentiate between a product business and a service business with one example each.
[2]

Expected answer / marking points: Product — sells goods (bakery, shoe shop) [1]; service — provides services (salon, tuition centre) [1].

Q9 **Question:** The stage of the AI Project Cycle in which the model is tested for accuracy is:
[1]

(a) Problem Scoping (b) Data Acquisition
(c) Modelling (d) Evaluation

Expected answer / marking points: (d) Evaluation

Q10 **Question:** Which AI domain helps a self-driving car "see" traffic signals?
[1]

(a) NLP (b) Computer Vision
(c) Statistical Data (d) Speech

Expected answer / marking points: (b) Computer Vision

Q11 **Question:** The "Who" block of the 4Ws canvas identifies the:
[1]

(a) stakeholders (b) location
(c) reasons (d) data

Expected answer / marking points: (a) stakeholders

Q12 **Question:** Data that cannot be easily arranged in rows and columns (like photos and videos) is:
[1]

(a) structured (b) unstructured
(c) numeric (d) ordinal

Expected answer / marking points: (b) unstructured

Q13 **Question:** Which chart is best to show the percentage share of each subject in study time?
[1]

(a) Line graph (b) Pie chart
(c) Scatter plot (d) Histogram

Expected answer / marking points: (b) Pie chart

Q14 **Question:** The median of 3, 8, 5, 10, 7 is:
[1]

(a) 5 (b) 7
(c) 8 (d) 6.6

Expected answer / marking points: (b) 7

Q15 **Question:** A die is rolled. The probability of getting a number less than 3 is:
[1]

(a) 1/6 (b) 1/3
(c) 1/2 (d) 2/3

Expected answer / marking points: (b) 1/3

Q16 [1] **Question:** Which of these is an example of Generative AI output?

(a) Calculator result
(b) An image made from a text prompt
(c) A spreadsheet total
(d) A printed bill

Expected answer / marking points: (b) An image made from a text prompt

Q17 [1] **Question:** What is the output of print(10 // 3) in Python?

(a) 3.33
(b) 3
(c) 1
(d) 30

Expected answer / marking points: (b) 3

Q18 [1] **Question:** Which of these is a valid Python variable name?

(a) 2marks
(b) total marks
(c) total_marks
(d) class

Expected answer / marking points: (c) total_marks

Q19 [2] **Question:** What is AI bias? Give one example.

Expected answer / marking points: Unfair results from an AI system because of one-sided/incomplete training data [1]; e.g., face recognition working poorly for some skin tones, or a hiring tool preferring one group [1].

Q20 [2] **Question:** Name any two sources of data for an AI project and one reason why data must be reliable.

Expected answer / marking points: Surveys, sensors/cameras, websites/APIs, government open data (any two) [1]; wrong data gives wrong predictions [1].

Q21 [2] **Question:** Find the mean and mode of: 4, 6, 4, 8, 10, 4, 6.

Expected answer / marking points: Mean = 42/7 = 6 [1]; mode = 4 [1].

Q22 [2] **Question:** Write two responsible ways to use a Generative AI tool for schoolwork.

Expected answer / marking points: Use it to understand/brainstorm, not to copy answers; check facts from reliable sources; mention when AI was used; don't share personal data (any two) [2].

Q23 [2] **Question:** Predict the output:

```
a = 5
b = 2
print(a * b, a % b)
```

Expected answer / marking points: 10 1 [1 + 1].

Q24 [4] **Question:** Explain the stages of the AI Project Cycle with the example of an app that predicts which school buses will be late.

Expected answer / marking points: Problem Scoping — who is affected, why delays matter [0.75]; Data Acquisition — GPS, traffic, weather, past timings [0.75]; Data Exploration — graphs of delays vs time/route [0.75]; Modelling — rule/ML model to predict delay [0.75]; Evaluation — test with new data, measure accuracy [0.5]; Deployment — alerts to parents [0.5].

Q25 **[4]**

Question: Explain the three domains of AI with two real-life examples of each.

ATHVA / OR

What is data literacy? Explain the steps of turning raw data into useful information using a school example.

Expected answer / marking points: Statistical Data — recommendations, price prediction [1.33]; Computer Vision — face unlock, traffic cameras [1.33]; NLP — voice assistants, translation, chatbots [1.33]. **OR:** Ability to read, understand, analyse and communicate data [1]; collect (marks of all students), clean (remove errors), organise (table), analyse (averages, graphs), interpret and decide (extra classes) [3].

Q26 **[4]**

Question: Write a Python program that asks for the marks of 3 subjects, calculates the total and average, and prints "Pass" if the average is 33 or more, otherwise "Fail".

Expected answer / marking points: Three input() with int/float [1]; total and average [1]; if–else condition [1.5]; print statements [0.5].

Q27 **[4]**

Question:

> A school canteen wants to reduce food waste. It records the number of plates of each dish prepared and sold every day for a month.

(i) Which stage of the AI Project Cycle is this? [1]
(ii) Name one more piece of data that would be useful. [1]
(iii) Suggest a graph to show daily waste of each dish and one decision the canteen could take from it. [2]

Expected answer / marking points: (i) Data Acquisition [1]. (ii) Weather, day of week, exam days, number of students present (any one) [1]. (iii) Bar graph/line graph [1]; prepare fewer plates of dishes with high waste on certain days [1].

Q28 **[4]**

Question:

> Riya asked a Generative AI chatbot to write her history project. The answer looked correct, but two dates were wrong and some sentences matched a website word for word.

(i) What problem with Generative AI does the wrong date show? [1]
(ii) What problem does the copied text show? [1]
(iii) How should Riya use the tool properly for her project? Give two points. [2]

Expected answer / marking points: (i) It can give wrong or made-up information ("hallucination") [1]. (ii) Plagiarism/copyright issue [1]. (iii) Use it for ideas/outline, verify facts from textbooks/reliable sites, write in her own words, cite sources [2].

Score: ___ / 50 Checked by: _____

Weak topic aur correction:

Parent Sunday handover checklist

- [ ] Agle Monday–Sunday ka student pack print karke diya.
- [ ] Parent solved key alag rakhi; student copy ke saath share nahi ki.
- [ ] Pichhle week ka pending kaam aur 3 weak topics review kiye.
- [ ] School exam date sheet aur tuition timings se schedule match kiya.

Agli priority / teacher se poochhna: